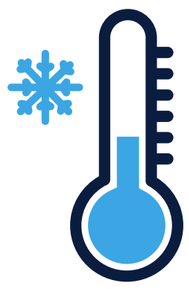
DATA TYPES

NUMERIC

Numeric data consists of continuous or discrete numbers that represent quantities, measurements, or counts.

Examples:

- Temperature readings
- Prices and sales figures
- Sensor outputs
- Population counts



TABULAR

Structured data organized into rows and columns, where each row is an instance and each column is a feature.

Examples:

- Spreadsheets
- Databases
- Sensor logs

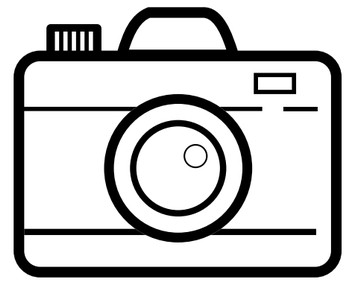


IMAGES

Image data consists of pixel grids representing visual content.

Examples:

- Medical scans (e.g., X-rays)
- Photographs
- Satellite imagery

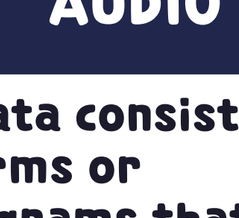


AUDIO

Audio data consists of waveforms or spectrograms that capture sound over time.

Examples:

- Speech recordings
- Music
- Environmental sounds
- Voice Commands



TIME SERIES

Time series data includes sequences of values indexed by time.

Examples:

- Stock prices over time
- Heart rate over time
- Weather data
- Website traffic logs



LEARNING PARADIGMS

SUPERVISED LEARNING

Learns from labeled input-output pairs.

Pros: High accuracy with enough data

Cons: Needs lots of labeled examples

Considerations:

Performance depends on label quality and representativeness

UNSUPERVISED LEARNING

Finds patterns in data without labels

Pros: Works with unlabeled data

Cons: Hard to evaluate results

Considerations:

Interpretability and validation can be difficult

REINFORCEMENT LEARNING

Learns through trial and error to maximize reward

Pros: Can learn complex, long-term behaviors

Cons: Data inefficient, unstable training

Considerations:

Requires simulation or safe interaction environment

SEMI-SUPERVISED LEARNING

Uses a small labeled set plus large unlabeled data

Pros: Reduces labeling cost

Cons: Can amplify label errors

Considerations:

Needs assumptions about data distribution

IMITATION LEARNING

Learns by mimicking expert demonstrations

Pros: Simpler to implement

Cons: Hard to evaluate results

Considerations:

Needs high-quality, diverse demonstrations

HUMAN-IN-THE LOOP LEARNING

Uses human feedback to guide learning

Pros: Increases accuracy and trust

Cons: Can be slow or costly

Considerations:

Depends on quality and consistency of human input

MODEL TYPES/TECHNIQUES

GRAPH NEURAL NETWORKS

Neural networks designed to learn from graph-structured data

Pros: Captures relationships and structure in data

Cons: Computationally intensive, harder to interpret

Considerations:

Requires graph input; performance sensitive to connectivity patterns

TRANSFORMERS

Attention-based architecture for sequence modeling

Pros: Highly flexible and scalable

Cons: Resource-heavy, needs lots of data

Considerations:

May struggle with small data or real-time use without tuning

BANDITS

Algorithms for sequential decision making under uncertainty

Pros: Balances exploration vs. exploitation

Cons: Can overfit to early observations

Considerations:

Performance depends on reward signal quality

CLUSTERING

Technique that groups similar data points together

Pros: Reveals hidden structure or patterns

Cons: No ground truth to validate against; interpretation can be subjective

Considerations:

May require tuning or pre-processing

RULE-BASED SYSTEMS

Uses hand-crafted rules to make decisions

Pros: Transparent and predictable

Cons: Inflexible and hard to scale

Considerations:

Requires expert-defined rules; brittle to unexpected input

DIFFUSION MODELS

Generative models that iteratively refine noise into data (e.g., images, audio)

Pros: Produce high-quality, diverse outputs; stable training dynamics

Cons: Slow inference speed due to iterative denoising steps

Considerations:

Requires significant compute for training and sampling

SYSTEM BEHAVIOR/SETTING

COMPLEMENTARY SYSTEMS

AI primarily functions independently but defers to a human or abstains when uncertain

Pros: Efficient for high confidence tasks, maintains human oversight

Cons: Limited collaboration — human input is only reactive

Considerations:

Requires well-calibrated uncertainty detection and division of responsibility

COOPERATIVE SYSTEMS

AI actively seeks additional input from humans when confidence is low

Pros: Proactively incorporates human insight

Cons: Can introduce inefficiencies if humans are over-relied on or fatigued

Considerations:

Human input must be relevant, available, and integrated seamlessly

FULLY INTERACTIVE SYSTEMS

Humans and AI engage in continuous, bidirectional interaction

Pros: Highly adaptive and responsive to changing user goals

Cons: Greater risk of unintended adaptation to reinforce biases, etc.

Considerations:

Must manage evolving goals, preferences, and system updates

DECISION SUPPORT SYSTEMS

Systems that analyze data to provide recommendations for human decision-making

Pros: Keeps human in control; improves outcomes with expertise

Cons: Over-trusting system outputs can lead to automation bias

Considerations:

Must present outputs clearly and justify them

AUTONOMOUS SYSTEMS

Operate independently in real-world settings with minimal human input

Pros: Scales without constant human input

Cons: Difficult to ensure safety in unpredictable scenarios

Considerations:

Must handle edge cases and make decisions under uncertainty

TRAINING REGIME

TRANSFER LEARNING

Adapts a model trained on one task or domain to a new but related one

Pros: Reduces training time and data needs

Cons: May fail if source and target tasks are too different

Considerations:

Pretrained model must align with target task

FEW-SHOT/META-LEARNING

Learns to perform new tasks with very few labeled examples

Pros: Fast adaptation to new tasks

Cons: Often hard to train and generalize

Considerations:

Needs well-defined task structure and diversity in training

PRE-TRAINED MODELS

Uses a model already trained on a general task and applies it to new tasks

Pros: Saves time and resources

Cons: May encode unknown biases from pretraining

Considerations:

Limited control over internal representations and behavior

TRAINED FROM SCRATCH

Models initialized randomly and trained on raw data without prior knowledge

Pros: Avoids biases from pre-trained models; domain-specific optimization

Cons: Risk of poor generalization if data is limited or noisy

Considerations:

Requires massive datasets and compute resources, time